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MILITARY SPECIFICATION

SPAGHETTI WITH MEATBALLS IN SAUCE, THERMOSTABILIZED, TRAY PACK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers spaghetti with meatballs in sauce, thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-C-495	- Coatings, Exterior, for Tinned Food Cans
PPP-B-636	- Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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MILITARY

MIL-L-1497	- Labeling of Metal Cans for Subsistence Items
MIL-L-35078	- Loads, Unit: Preparation of Semiperishable Subsistence Items; Personal Equipment and Equipage; General Specification For
MIL-C-44340	- Can, Tray Pack

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-900	- Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies may be obtained from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Institutional Meat Purchase Specifications for Fresh Beef, Series 100

(Copies may be obtained from the Director, Livestock and Seed Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2092, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

Meat and Poultry Inspection Regulations

(Copies may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

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U.S. Standards for Condition of Food Containers

(Copies may be obtained from the Chairman, Condition of Food Container Committee, Agricultural Marketing Service, U.S. Department of Agriculture, Room 0610, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. Standards for Grades of Tomato Paste

(Copies may be obtained from the Director, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2077, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Copies should be obtained from the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Copies should be obtained from the American Dehydrated Onion and Garlic Association, 650 California Street, Suite 800, San Francisco, CA 94108.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3330 - Peel Adhesion of Pressure-sensitive Tape

(Copies should be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Copies should be obtained from the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Copies should be obtained from the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.1, and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Beef. The beef shall be from steers, heifers, or cows and shall be derived from any combination of carcasses, quarters, and any of the following recognizable primal and subprimal cuts or portions thereof: square-cut chucks, shoulder clods, chuck rolls, ribs, ribeyes, trimmed full loins, trimmed short loins, strip loins, sirloins, top sirloin butts, rounds, top rounds, peeled knuckles, outside rounds, and bottom rounds (heel out).

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Recognizable cuts are those which, when compared to Institutional Meat Purchase Specifications (IMPS) cuts, have no more than a minor amount of lean, fat, or bone removed or included from an adjacent cut. A portion is a certified recognizable cut from which objectionable tissue, such as tendons or heavy connective tissue, has been removed. Tenderloins, rib fingers (intercostal meat), and any part of a rough cut may be excluded at the option of the contractor. The beef shall be in the fresh-chilled state and shall be in excellent condition; i.e., exposed lean and fat surfaces shall be of a color and bloom normally associated with the class and cut of meat, and typical of meat which has beeen properly stored and handled. Cut surfaces and naturally exposed lean surfaces shall show no more than slight darkening or discoloration due to dehydration, aging, or microbial activity. The fat shall show no more than a very slight discoloration due to oxidation or microbial activity. No odors foreign to fresh meat shall be present. Changes in color and odor characteristically associated with vacuum-packaged meat in excellent condition shall be acceptable. Also, the meat shall show no evidence of freezing, defrosting, or mishandling.

3.2.1.1 Boning and trimming. The beef shall be boned and trimmed to remove objectionable material such as bone, cartilage, heavy connective tissue, etc. The boneless trimmed beef shall meet the limitations specified in tables I and II.

3.2.1.2 Handling and storage. Handling and storage of the boned and trimmed beef prior to processing into the finished product shall be in accordance with the following requirements:

a. Beef processed on the day of initial certification shall be maintained in the temperature range of 28° to 50°F (inclusive).

b. Holding in the fresh chilled state for not more than 4 days after initial certification is permitted provided that the beef is maintained in the temperature range of 28° to 40°F (inclusive).

c. Holding in the frozen state for not more than 120 days after initial certification is permitted provided that the beef is:

- frozen to 0°F or lower within 72 hours after initial certification
- stored at 0°F or lower
- protected from freezer deterioration and damage
- stored in containers that are adequate to maintain product excellence
- held after storage at an internal temperature not to exceed 40°F when further processing is resumed.

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3.2.2 Meatballs, cooked, frozen. Meatballs received in a cooked, frozen state shall be accompanied by a USDA Agricultural Marketing Service (AMS) Certificate certifying the requirements of 3.2.1, 3.2.1.1, 3.2.1.2, 3.3.1, and 3.3.2 have been met. The cooked meatballs shall have been vacuum packaged and heat sealed within 4 hours after the cooking process. The packaging material shall be USDA approved moisture and oxygen impermeable material. Within 72 hours after packaging, the packaged meatballs shall have been frozen to an internal temperature of 0°F or below. The cooked packaged, frozen meatballs shall have been held at an internal temperature of 0°F or lower for not more than 60 days prior to tray pack filling.

3.2.3 Bread crumbs. Bread crumbs shall be made from white bread only. It shall be coarse ground, free-flowing and free from burnt, rancid, moldy, or musty odors or flavors.

3.2.4 Salt. Salt shall be noniodized, white, refined, sodium chloride with or without anticaking agents and shall comply with the purity standards for sodium chloride of the Food Chemicals Codex.

3.2.5 Pepper, black, ground. The ground black pepper shall have been ground from the deep brown to black, deep-set, wrinkled, immature berries of Piper nigrum L. The ground pepper shall have a characteristic, penetrating odor, a hot biting pungent flavor, and a light gray to speckled black-gray color. The black pepper shall contain not less than 2.0 mL of volatile oil per 100 grams and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 16 sieve.

3.2.6 Onions, dehydrated, chopped. The dehydrated chopped onions shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.7 Tomato paste. Tomato paste shall be U.S. grade A of the U.S. Standards for Grades of Tomato Paste and shall be of the latest season's pack.

3.2.8 Water. Water used for formulation, ice making, and washing shall conform to the National Primary Drinking Water Regulations.

3.2.9 Tomatoes, crushed, canned. Canned tomatoes shall be peeled, cored, mature, crushed tomatoes. The use of safe and suitable firming and acidification ingredients and salt is permitted. The canned tomatoes shall have not less than 8.0 percent tomato soluble solids and shall possess a red flesh color, normal character, and a good distinct acid sweet tomato flavor and odor. The crushed tomatoes shall be free of extraneous vegetable material and objectionable core material and skins (peel). The canned, crushed tomatoes shall be of the latest season's pack.

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3.2.10 Sugar, white, granulated. Sugar shall be white, refined, sucrose, granulated cane or beet or a combination thereof and shall comply with MIL-STD-900.

3.2.11 Starch, food, modified, high opacity. The high opacity starch shall be a white, odorless, finely pulverized, modified maize food starch for use in thermostabilized foods and shall comply with MIL-STD-900. The modified starch shall demonstrate initial viscosity development in the temperature range of 140° to 170°F and a typical viscosity (be fully hydrated) at common retort temperatures. The starch shall resist breakdown at low pH, under shear stress, and under conditions of cold storage. The cooked starch shall be bland with essentially no cereal or starch taste.

3.2.12 Oregano, crushed or course ground. Oregano shall be derived from the dried leaves of Origanum vulgare L. and shall have a strong camphoraceous aroma and a pungent slightly bitter flavor. Volatile oil content shall be not less than 2.0 mL/per 100 grams and the oregano shall be of such size that not less than 95 percent shall pass through a U.S. Standard No. 20 sieve.

3.2.13 Garlic powder. Garlic powder shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.14 Basil, ground. Ground basil shall be derived from the dried leaves of Ocimum basilicum L. and possess a sweet, anise-like odor, and aromatic, warm, slightly pungent flavor. A minimum of 95 percent by weight shall pass through a U.S. Standard No. 35 sieve. The volatile oil content shall be not less than 0.4 mL/per 100 grams.

3.2.15 Fennel, ground. Ground fennel shall be derived from the clean ripe fruit of Foeniculum vulgare Mill. The ground fennel shall be light green-brown to yellow-brown in color and possess a pleasant aromatic odor with a flavor resembling that of anise. A minimum of 95 percent by weight shall pass through a U.S. Standard No. 25 sieve. The volatile oil content shall be not less than 1.0 mL/per 100 grams.

3.2.16 Pepper, red ground. Ground red pepper shall be derived from red, ripe fruit of the genus Capsicum frutescens L. and shall possess the characteristic red to brown-red color with a perceptible sensation of pungency. It shall be uniformly ground to allow for a minimum of 95 percent by weight to pass through a U.S. Standard No. 40 sieve and not less than 95 percent by weight to be retained on a U.S. Standard No. 60 sieve. The heat value shall be not less than 30,000 Scoville Heat Units.

3.2.17 Spaghetti (thin). The spaghetti shall be enriched, shall have a 0.06 to 0.083 inch diameter, and shall be the solid cylindrical type. The spaghetti shall be made from semolina flour only and shall contain distilled glycerol-monostearate. Spaghetti shall be cut into approximately 2-inch lengths.

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3.2.18 Fat, frying. Frying fat shall be prepared from animal or vegetable (excluding coconut oil and palm kernel oil) fats, or a blend of animal and vegetable fats, which have been hydrogenated and deodorized. The frying fat shall have a clean bland flavor. The frying fat may be liquid or plasticized, may contain antioxidants in combination and amounts allowed by FDA regulations, and shall contain antifoaming agents. The frying fat shall have a maximum free fatty acid value of 0.05 percent (expressed as oleic acid), a stability of not less than 40 hours (A.O.M.) and a smoke point of not less than 400°F.

3.2.19 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this document. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the blend. The ingredients shall be in the same proportions as specified in the ingredient formula.

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Beef preparation. The beef shall be prepared as follows:

a. Boned and trimmed beef meeting the requirements of 3.2.1.1 shall be further trimmed, if necessary, so as to ensure compliance with finished product requirements (see 3.6).

b. The beef shall be mechanically ground through a plate having holes of 1 inch diameter. After grinding, the beef shall be maintained for not more than 24 hours at an internal temperature of 28° to 40°F prior to meatball preparation.

3.3.2 Meatball preparation. The meatballs shall be formulated and prepared as follows:

Ingredients	Percent by weight
Beef (raw basis)	91.71
Bread crumbs	3.76
Salt	1.00
Black pepper	.03
Onions, rehydrated	2.00
Tomato paste	1.50
	100.00

a. The ingredients shall be combined and thoroughly mixed. The meatball mixture shall be ground through a plate having holes 3/16 inch in diameter or chopped to give equivalent results and formed into meatballs of uniform weight and size to comply with finished product drained weight requirements. The meatball mixture and the meatballs shall be maintained at an internal temperature of 28° to 40°F for not more than 12 hours prior to the browning (cooking) process.

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b. The meatballs shall be surface browned until no pink color is visible in the interior of the meatball. The meatballs may be cooked by any commercial method except moist heat (steaming). Frying fat, if used, shall comply with requirements of 3.2.18. The fat shall be filtered at least daily and the frying system cleaned at least weekly. The fat shall be discarded when foaming, undesirable odors and flavors or other evidence of fat breakdown is apparent. If the meatballs are not filled into the tray pack immediately after cooking, they shall be held for not more than 48 hours at an internal temperature of 28° to 40°F prior to filling and sealing of the tray pack can.

c. The cooked meatballs may be stored in the frozen form provided that within 4 hours after the cooking process, the meatballs are vacuum packaged and heat sealed in packaging material complying with the requirements specified in 3.2.2. The meatballs shall be frozen to an internal temperature of 0°F or below within 72 hours after packaging. The packaged, frozen, cooked meatballs shall be maintained at an internal temperature of 0°F or below for not more than 60 days prior to tray pack filling.

3.3.3 Spaghetti preparation. The spaghetti shall be blanched in boiling water until doubled in weight. The spaghetti shall be immediately rinsed in cold water to reduce the temperature to 90°F or lower and to remove excess starch. If the spaghetti is not immediately combined with the sauce, it shall be cooled to 45°F or lower and held for not more than 4 hours prior to combining with the sauce.

3.3.4 Sauce preparation. The sauce shall be formulated and prepared as follows:

Ingredients	Percent by weight
Water	63.19
Crushed tomatoes (8.0 percent solids) 1/	23.85
Tomato paste (32 percent solids) 1/	8.60
Sugar	1.25
Salt 2/	1.25
Onions, dehydrated	0.87
Starch, modified	0.80
Oregano, crushed or coarse ground	0.06
Garlic powder	0.05
Basil, ground	0.04
Red pepper, ground	0.03
Fennel, ground	0.01

1/ If the specified tomato solids are not available, the tomato solids that are used must be calculated to bring the solids level to that specified.

2/ The total amount of salt in the sauce formula shall be adjusted as necessary to produce a product that complies with the finished product salt requirements (see 3.6r).

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NOTE: The following sauce preparation procedures were used in the development of this product. Alternative procedures may be used provided finished product requirements are met. (When alternative procedures are used, the time and temperature requirements specified for the prepared sauce are still applicable.)

a. The starch and part of the water shall be mixed to form a slurry.

b. The balance of the water shall be heated in a steam-jacketed kettle until the temperature reaches 180° to 190°F.

c. The starch slurry shall be added to the water and the mixture stirred continuously until thickened and the temperature reaches 180° to 190°F.

d. The steam supply to the kettle shall be shut off and the balance of ingredients added and uniformly mixed to form the sauce.

e. The volume of the sauce shall be adjusted with water prior to filling to compensate for evaporation loss during heating and holding.

f. The sauce shall be maintained in the temperature range of 150° to 180°F for not more than 4 hours prior to filling into tray pack cans.

3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with meatballs, spaghetti, and sauce such as to conform to the finished product requirements and to the following requirements:

a. Each can shall be hermetically sealed under a vacuum established by a processing authority and specified in the scheduled process so as to assure compliance with the finished product requirement (see 3.6t).

b. The filled and sealed tray pack cans shall be in the retort process within 2 hours after sealing.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting until a sterilization value (F_o) of not less than 6 has been achieved.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

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d. Total weight of cartilage, coarse connective tissue, section of tendons or ligaments, and glandular material, collectively, in a can shall be not greater than 1 ounce.

e. No individual can shall contain a bone piece measuring 0.3 inch or more in any dimension.

f. Meatballs shall be uniform in size and shape.

g. Meatballs texture shall not be dry, tough, rubbery, or mushy.

h. Spaghetti texture shall be firm, not mushy or tough.

i. Sauce consistency shall be smooth, and shall not be lumpy or excessively thick or thin.

j. Each can shall contain a minimum of 24 meatballs.

k. The drained weight of 24 meatballs in a can shall be not less than 22.0 ounces.

l. The average drained weight of the meatballs shall be not less than 24.0 ounces.

m. No individual can shall contain less than 50.0 ounces or more than 60.0 ounces drained weight of spaghetti.

n. The average net weight shall be not less than 106 ounces.

o. No individual can shall have a net weight of less than 104 ounces of product.

p. The average fat content of the finished product shall be not greater than 10.0 percent.

q. No individual tray pack shall have a fat content greater than 12.0 percent.

r. No individual tray pack shall have a salt content greater than 1.3 percent or less than 0.5 percent.

s. The product shall show no evidence of excessive heating (materially darkened or scorched).

t. Filled, sealed, and retorted cans shall show evidence of proper vacuum as determined by concavity of the can lid (see 4.5.6).

3.6.1 Palatability. The finished product shall be equal to or better than the preproduction sample (see 6.1) in palatability and overall appearance.

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3.7 Plant qualification. The beef component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all the requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection necessary to assure compliance with the requirements of this document.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Beef examination for condition and cut. All beef shall be examined in either the bone-in or boneless state for conformance to the condition and cut requirements in 3.2.1. Cuts initially examined in the boneless state shall be in the form of whole boneless recognizable cuts or portions thereof. Any nonconforming beef shall be rejected.

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4.5.1.2 Boned and trimmed beef examination. After boning and trimming and prior to any further processing or to any freezing, the beef shall be examined for the defects listed in table II. The lot size, expressed in terms of pounds, shall be declared to the Agricultural Marketing Service (AMS) agent by the contractor. However, the AMS agent reserves the right to declare as a lot a portion of a declared lot, if, in his or her opinion, that portion may be out of compliance with any requirement. The sample unit shall be a minimum of 12 pounds of adjacent boneless beef. The sample size shall be as specified in table I. If all or a portion of the sample unit falls within a larger cut, the entire cut shall be examined. Failure of the beef to meet the acceptance criteria as indicated in table I shall be cause for rejection of the lot. Except for beef rejected because of freezing, defrosting, or not being in excellent condition, the beef may be reworked by the contractor and reoffered for examination. For reexamination, the sampling plan used shall be the one in table I designed for the next larger lot size than the one under which the lot was initially rejected. Beef shall not be reexamined more than one time.

TABLE I. Sampling plan for boned and trimmed beef

Lot size (pounds)	Sample size (sample units)	Defect categories			
		Major		Minor	
		AC	RE	AC	RE
500 or less	20	2	3	5	6
501 to 1,200	32	3	4	7	8
1,201 to 3,200	50	5	6	10	11
3,201 to 10,000	80	7	8	14	15
10,001 to 38,000	125	10	11	21	22
Reinspection lots of 10,001 to 38,000	200	14	15	21	22

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TABLE II. Boned and trimmed beef defects 1/ 2/

Category		Defect
Major	Minor	
101		Presence of popliteal, prescapular, prefemoral or any exposed lymph gland measuring 0.5 inch or more in any dimension
102		Presence of blood clot measuring 1.0 inch or more in any dimension
	201	Presence of blood clot measuring 0.5 inch or more but less than 1.0 inch in any dimension
103		Presence of bruise measuring 1.0 inch or more in any dimension
	202	Presence of bruise measuring 0.5 inch or more but less than 1.0 inch in any dimension
104		Presence of bone measuring 0.3 inch or more in any dimension
105		Presence of cartilage measuring 0.5 inch or more in any dimension
106		Presence of backstrap measuring 1.0 inch or more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other)
	203	Presence of backstrap measuring 0.5 inch or more but less than 1.0 inch in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other)
107		Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin, or the gracilis membrane) measuring 2.0 square inches or more
	204	Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin, or gracilis membrane) measuring 0.5 square inch or more but less than 2.0 square inches

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TABLE II. Boned and trimmed beef defects 1/ 2/ (cont'd)

Category		Defect
Major	Minor	
	205	Presence of heavy connective tissue on lower edge of short plate or flank
	206	Presence of heavy connective (abdominal tunic) tissue on the flank measuring 1.0 inch or more in any dimension
	207	Presence of membranous portion of diaphragm or membranous covering from skirt, flank, or abdominal section of short plate measuring 3.0 square inches or more
108		Presence of knuckle cover
109		Presence of kidney, pizzle eye, prepubic tendon, thymus gland, or hanging tender measuring 1.0 inch or more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other)
	208	Presence of kidney, pizzle eye, prepubic tendon, thymus gland, or hanging tender measuring less than 1.0 inch in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other)
	209	Presence of calcified (scratchy) periosteum measuring 2.0 square inches or more
	210	Presence of shank, clod, knuckle, or bottom (outside) round with tendinous end showing less than 75 percent lean tissue on a cross-sectional cut surface
	211	Presence of dehydrated surface measuring 1.0 square inch or more
	212	Presence of discolored meat (including blood discolored neck meat) measuring 1.0 square inch or more
	213	Presence of exposed blood vessel measuring 1.0 inch or more in any dimension
	214	Presence of cod, udder, kidney, or pelvic fat

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1/ Determination of wholesomeness and acceptability of product with respect to the presence of foreign material (e.g., glass, dirt, insect parts, hair, wood, metal) shall be made by a Meat and Poultry Inspection Operations employee.

2/ Evidence of freezing or defrosting or product not in excellent condition shall cause rejection of the lot.

4.5.1.3 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.5.3.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1 ounce. If the average net weight is less than 106 ounces or if the net weight of any individual can is less than 104 ounces, the lot shall be rejected.

4.5.3.2 Product inspection. The sample size shall be as indicated by the double sampling plan specified in table III. The tray pack cans shall be selected at random from the lot. The tray pack cans shall be heated for 35 minutes in boiling water, opened, and inspected for the defects listed in table IV.

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TABLE III. Double sampling plan for product inspection 1/

Lot size (cans)	Sample size (cans)	Cumulative sample	Acceptance number	Rejection number
0 to 3,200	8	-	0	2
	8	16	1	2
3,201 to 35,000	13	-	0	3
	13	26	3	4

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second samples shall be combined and, if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE IV. Product defects 1/ 2/

Category	Defect
Major	
101	Less than 24 meatballs in a can
102	Drained weight of 24 meatballs in a can is less than 22.0 ounces 3/ 4/
103	Drained weight of spaghetti in a can is less than 50.0 ounces or more than 60.0 ounces 3/
104	Total weight of cartilage, coarse connective tissue, section of tendons or ligaments, and glandular material, collectively, in a can is greater than 1 ounce

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TABLE IV. Product defects 1/ 2/ - Continued

Category	Defect
Major	
105	Presence of bone measuring 0.3 inch or more in any dimension
106	Meatballs not uniform in size and shape
107	Meatballs dry, tough, rubbery, or mushy
108	Spaghetti is hard or mushy
109	Sauce consistency is not smooth, is lumpy, or is excessively thick or thin
110	Product shows evidence of excessive heating (materially darkened or scorched)

1/ The presence of foreign material (e.g., glass, dirt, insect parts, hair, wood, metal), foreign odor or flavor (e.g., burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an Agricultural Marketing Service (AMS) agent.)

3/ To determine drained weight, the free liquid in the can shall be poured off and the remaining contents shall be poured into a flat bottom container. A minimum of three times the tray pack can's volume of 190° to 212°F water shall be added to the container so as to cover the contents. The contents and water shall be agitated so as to liquify rendered fat and to remove the sauce without breaking the meatballs or spaghetti. The contents shall then be poured into a U.S. Standard 1/4-inch sieve in a manner that will distribute the product over the sieve without breaking the meatballs or spaghetti. Sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted at an approximate 45-degree angle and allowed to drain for 2 minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. After gross weight has been attained, remove meatballs and reweigh. The drained weight shall be reported to the nearest 0.1 ounce.

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4/ The lot shall be rejected if the sample average drained weight of the meatballs is less than 24.0 ounces.

4.5.3.3 Fat and salt content testing. Nine tray pack cans shall be selected at random from the lot and distributed as follows:

- Three for laboratory analysis.
- Three for submission to the contractor.
- Three for retention by the AMS agent as reserve samples.

The three tray packs for laboratory analyses shall be individually tested for fat and salt content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, chapter: Meat and Meat Products, except that preparation of the samples shall be as follows. The three unopened cans shall be gently warmed in a water bath to melt fat adhering to the inside of the cans. The cans shall be opened and the entire contents of each can shall be separately blended in a Waring blender or equivalent. The test results shall be reported to the nearest 0.1 percent. Any test failure is a major defect. The lot shall be rejected if any of the following major defects occur:

a. The average fat content of the three tray packs is greater than 10.0 percent.

b. The fat content of any individual tray pack is greater than 12.0 percent.

c. The salt content of any individual tray pack is greater than 1.3 percent or less than 0.5 percent.

Analysis of reserve samples at the request of the contractor shall not be permitted unless the original laboratory analysis indicated that the involved lot will be rejected because of noncompliance with the fat or salt content requirement. When the reserve samples are analyzed, the analyses for both fat and salt shall be made and will be considered final. Unused reserve samples shall be returned to the contractor for inclusion in subsequent lots.

4.5.4 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers except that inspection for labeling shall be in accordance with 4.5.4.1. In addition, the following defect shall be classified as a critical defect and shall be cause for rejection of the lot:

Evidence of buffing causing damage (i.e. scratches or scuffing) to exterior can coating.

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4.5.4.1 Can label examination. Labels shall be examined for defects in accordance with MIL-L-1497 (see 5.4), except that for self-adhering labels the following additional defects shall apply:

Major: Labels torn or scratched so as to obliterate any of the markings.

Minor: Air bubbles under label.
Label not properly adhered to can, i.e, label raised or peeled back from edges or corners.

4.5.4.2 Label adhesive examination. When self-adhering labels are used, the adhesive shall be tested in accordance with ASTM D 3330.

4.5.5 Can closure examination. Can closures shall be examined visually and by teardowns in accordance with the can manufacturer's requirement and 21 CFR, Part 113, Subpart D, or 9 CFR, Part 318, Subpart G, as applicable. Any nonconformance based on observation of can seam teardowns or on record of can seam teardowns is a major defect and shall be cause for rejection of any involved product.

4.5.6 Vacuum examination. Cans shall be allowed to cool to 75° ± 5°F, held for at least 24 hours after sealing, and then examined for vacuum retention. To examine, lay a straight edge in the center of the lid along the length of the tray pack. Both ends of the straight edge shall touch the lid at the inside edge of the double seam. There shall be a visible gap between the straight edge and the lid for the entire distance of the label panel. Using a shorter straight edge, the same procedure shall be used across the width, in the center of the tray pack can. When examining a ribbed lid, lay the straight edge only between the two center ribs along the length of the can. The inspection lot shall include only tray packs produced in a single shift on a single sealing machine. The sample size shall be 50 cans. Any nonconformance is a major defect and shall be cause for rejection of the lot.

4.5.7 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible.
Reinforced with other than nonmetallic strapping or tape.
Dimensions of pads not as specified.
Interior packing with fiberboard liner or pads not as specified.

Minor: Other required markings missing, incorrect, or illegible.
Arrangement or number of cans not as specified.

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4.5.8 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PACKAGING

5.1 Preservation. The product shall be preserved in accordance with level A.

5.1.1 Level A. One hundred and six ounces of food product shall be filled into a tray pack can conforming to MIL-C-44340 and sealed and thermoprocessed as specified in 3.4 and 3.5. Buffing of the can causing damage to the exterior can coating is not permitted (see 4.5.4).

5.2 Packing. The product shall be packed in accordance with level A, B, or C, as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L or HSC-L with an HSC full depth cover, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box, with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads fabricated of grade V3c fiberboard. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). Flute direction of the box liner shall be vertical. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box. Each box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament-reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1), except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern so as to create columnar stacking. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Four cans of product, preserved as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.3 Level C packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box, with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full

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acceptance and safe delivery packed in a manner to insure carrier a

inside depth of the box (+0 inch, -1/8 inch). Flute direction of the box liner shall be vertical. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of the same material as the box.

5.3 Unit loading. When specified (see 6.1), the product, packed as specified in 5.2.2 or 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified, except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern so as to create columnar stacking. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497 and with the following:

- Official establishment number (e.g., est 38) or a three-letter code identifying the establishment
- Lot number 1/
- Production shift number 1/
- Retort identification number 1/
- Retort cook number 1/

1/ The lot number shall be expressed as a four-digit Julian code. The first digit shall indicate the year of production and the next three digits shall indicate the day of the year (Example, March 19, 1987, would be coded as 7078). The Julian code shall represent the day the product was packaged and processed. Sub-lotting (when used) shall be represented by an alpha character immediately following the four-digit Julian code. Following the four-digit Julian code and the alpha character (when used), the other required code information shall be printed in the sequence as listed above.

In addition, the name of product shall be marked, stamping is permitted, on one 1001 by 200 side of the can. The labeling shall be legible when examined as specified in 4.5.4 after preparation of the product in accordance with heating instructions. Paper labels are not permitted. Cans shall show the following statements:

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TO HEAT: Submerge unopened can in boiling water. Simmer gently 40 - 45 minutes. Avoid overheating (can shows evidence of bulging).

CAUTION: Use care when opening as pressure may have been generated within the can.

To heat in oven: Either punch several holes in lid of can or open can in usual manner leaving the loose lid in place. Place in a 350°F oven 35 - 40 minutes.

WARNING: Do not place unopened can in oven. This may cause the can to burst.

YIELD: Serves 12 portions of 1 cup and 2 meatballs each.

A preprinted self-adhering 0.002-inch thick, clear, polyester label printed with indelible black ink may be used. Self-adhering labels shall be applied after retorting. Pressure-sensitive adhesive shall require no preparation prior to application. Labels shall tack quickly and adhere without curling or breaking. The adhesive shall have a minimum adhesion of 60 ounces per inch width when examined as specified in 4.5.4.2. When self-adhering label is used, the tray pack can shall be labeled with the Julian code and a product code prior to retorting.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, the following precautionary marking in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT
OR MORE THAN TWO HIGH IN STORAGE,
UNLESS PALLET RACKS ARE USED.

6. NOTES

6.1 Ordering data. Acquistion documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.4, and 6.3).
c. Provisions for approved preproduction samples (see 3.6.1 and 6.3).
d. Level of packing required (see 5.2).
e. Type and class of unit load when unit loading is required (see 5.2.1 and 5.3).

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6.2 Appropriate level of pack. Based on conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Subject term (key word) listing.

Canned foods
Combat field feeding
Spaghetti with meatballs
Thermostabilized
Tray pack

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing activity:

Army - GL

Project No. 8940-0612

Review activities:

Army - MD, TS
Navy - MC
DP - SS

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submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-44341	Spaghetti With Meatballs In Sauce, Thermostabilized, Tray Pack

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 1 Apr 84

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